

Junyan Tan (Samuel)

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Los Angeles, CA, USA

RESEARCH INTERESTS

Population Genetics; Quantitative Genetics; Machine Learning.

RESEARCH EXPERIENCE

- University of Michigan, Department of Statistics** May 2025 – Jan 2026
PI: Prof. Jonathan Terhorst Ann Arbor, USA
 - Co-created **demesinfer** , a Python library for demographic inference; implemented a JAX-based optimizer that automates demographic parameter estimation (replacing manual visual calibration in earlier workflows). Co-author on the demesinfer paper.
 - Applied demesinfer to real datasets, including an internal unphased Michigan snake dataset, and participated in the Genomic History Inference Strategies Tournament (GHIST) using demesinfer for inferring demographic parameters.
- University of Michigan School of Public Health, Department of Biostatistics** Aug 2024 – May 2025
PI: Prof. Irina Gaynanova Ann Arbor, USA
 - Developed a zero-shot LLM framework **Gluco-LLM**  with Retrieval-Augmented Generation (RAG) for CGM time-series forecasting, achieving up to a 30% reduction in error and 50% lower compute time compared to strong time-series baselines.
 - Maintained and extended the **iglu**  R package for CGM analysis (bug fixes, new features, tests, documentation) and enhanced the Shiny-based visualization interface.
 - Curated and processed **Awesome-CGM** , a widely used public CGM dataset for 1,000+ individuals with over 200 stars on Github. Coordinated analyses for a multi-institution consensus study with five external CGM software teams.
- Academy of Mathematics and Systems Science, Chinese Academy of Sciences** May 2024 – Aug 2024
PI: Prof. Qizhai Li Beijing, China
 - Developed a novel test statistic for multimodal regression with an R implementation and theoretical guarantees.
 - Demonstrated $1.5\text{--}2\times$ higher statistical power on simulated settings compared to prior tests; prepared an internal report.

EDUCATION

- University of Southern California** Current
Ph.D. Computational Biology and Bioinformatics Los Angeles, USA
- University of Michigan** 2026
M.S. Biostatistics Ann Arbor, USA
- University of Michigan** 2025
B.S. Data Science & Statistics, Honors in Data Science Ann Arbor, USA

HONORS AND AWARDS

- Phi Beta Kappa** Apr 2025
Phi Beta Kappa - Alpha of Michigan Chapter 
- Best Poster Award** Mar 2025
Michigan Student Symposium for Interdisciplinary Statistical Sciences 

PUBLICATIONS, PREPRINTS & SELECTED WORKS

- [1] **Tan, J.** (2025). *Evaluating the Zero-Shot Predictive Ability of Large Language Models for Continuous Glucose Monitoring Data*. Undergraduate Honors Thesis, University of Michigan.
- [2] Dilber, E., Liang, J., **Tan, J.**, Terhorst, J. (in revision). *Faster inference of complex demographic models from large allele frequency spectra*.
- [3] Xu, X., Kok, N., **Tan, J.**, et al. (2024). *Awesome-CGM: List of CGM datasets*.

TALKS & PRESENTATIONS

- **Evaluating the Zero-Shot Predictive Ability of LLMs for CGM Time Series** Mar 2025
MSSISS 2025 — Poster (Best Poster Award) 
 - Presented honors-thesis results on zero-shot LLM forecasting for CGM time series; highlighted gains and the role of retrieval-augmented generation (RAG).

SKILLS

- **Programming Languages:** R, Python, C++, C, SQL, Java, JavaScript
- **Machine Learning & Stats:** LLM, CNN, MCMC, HMM, Time-Series Forecasting
- **Tools & Frameworks:** JAX, PyTorch, TensorFlow, Hugging Face, Git, GCP, HPC Computing
- **Languages:** English, Chinese, Japanese (fluent); French (fluent); Spanish (working knowledge)

MENTORSHIP & OUTREACH

- **Data Science Consultant** Jan 2024 – Apr 2024
Industrial Sewing and Innovation Center (ISAIC) 
 - Analytics engagement for a Detroit-based nonprofit: Python-based analyses and visualizations (seaborn, matplotlib) and a concise report used in grant applications.
 - *My contribution:* designed the analysis plan, created the visuals, and co-authored the report.
- **Education Committee** Jan 2024 – Aug 2024
Michigan Data Science Team (MDST) 
 - Programmed training content for student practitioners, emphasizing statistical foundations and theory in data science (regularization, cross-validation, Bayesian methods).
 - *My contribution:* led curriculum design and delivery and mentored the project team.